

Hyalen Moreira Caldeira

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Portfolios: hyalen.us & hyalen.us/resume

ABOUT

Senior Software Engineer specializing in **AI** applications and enterprise-scale systems. Currently building **multi-agent** platforms with **LangChain/LangGraph/MCP**, implementing **RAG** pipelines, and designing **agentic workflows** that solve real business problems. Proven expertise in **full-stack** development (**Java/Spring, Python**), **SAP BTP/CAP/CPI** integration, and **DevOps/SRE** automation. Strong track record architecting scalable, production-ready solutions across **AI** systems, **cloud-native** applications, and enterprise integrations.

PROFESSIONAL EXPERIENCE

Accenture – as a Full Stack Engineering Manager & AI Architect *Columbus/OH, Jan/2023 – Current*

Leading enterprise AI platforms, multi-agent systems, and code generation solutions. Architect and deliver end-to-end AI systems, MCP platforms, and LLM-driven workflows from concept to production. Lead cross-functional teams, mentor engineers, and drive technical excellence across client engagements.

Key Projects & Contributions:

- **Multi-Agent Code Generation Platform** – Built an AI-powered system using Python, LangGraph, LangChain, and MCP that automates software creation through dynamic workflow routing. Clients configure agents, prompts, tasks, and routing via JSON, reducing development time by ~30% and significantly lowering manual rework.
- **Model Context Protocol (MCP) Blueprint** – Architected a company-wide MCP foundation used across multiple teams and clients. Standardized AI application architecture with support for SSE, HTTP, and STDIO transports, accelerating new project setup by ~40%.
- **BPMN Workflow Documentation Tool** – Developed an AI solution that reverse-engineers XML-based BPMN workflows for cross-application integration, auto-generating comprehensive documentation, user stories, and development artifacts.
- **AI Engineering Standards** – Established prompt engineering patterns, context management strategies, and design principles for multi-agent orchestration, enabling modular, maintainable LLM-driven workflows.

Impact: Transformed the company's AI capabilities from proof-of-concept to production-ready platforms. Mentored engineers, established code review practices that improved solution quality by 35%, and created reusable frameworks that accelerated delivery across client projects.

Accenture – as a Full Stack Engineering Associate Manager & DevOps/SRE & SAP BTP/CAP/CPI *Mar/2019 – Dec/2022*

Led full-stack development and SRE initiatives across multiple enterprise clients, delivering scalable cloud platforms, self-healing infrastructure, and SAP integrations. Designed resilient architectures, implemented automation workflows, and mentored engineering teams to drive high-quality delivery.

Key Projects & Contributions:

- **Self-Healing Infrastructure Platform** – Built an automated recovery system using Java/Spring Boot and Angular with Grafana/Splunk for real-time monitoring, reducing incident response time by ~45% and minimizing manual intervention during production outages.
- **Kubernetes Optimization (AKS/EKS)** – Enhanced cluster reliability through optimized configurations, blue-green rollout strategies, and deployment automation, achieving improved uptime and operational efficiency.
- **Infrastructure as Code** – Developed Terraform-based IaC modules to automate provisioning of AWS services (EC2, EKS, VPC, security groups, load balancers), cutting environment setup time from days to hours.
- **SAP BTP Integration & Event-Driven Architecture** – Designed CAP/CPI solutions for real-time data ingestion, transformation, and analysis, enabling seamless enterprise system integration and event processing.
- **CI/CD Pipeline Engineering** – Built GitLab pipelines automating builds, tests, and deployments across environments, increasing delivery speed and reducing production defects.
- **Observability & Monitoring** – Implemented Prometheus metrics, ELK stack logging, and distributed tracing to accelerate incident detection and streamline root-cause analysis, reducing MTTR by ~40%.

Impact: Delivered production-grade DevOps patterns and end-to-end observability across AWS and SAP BTP platforms. Enabled faster release cycles, reduced system downtime, and established SRE best practices adopted across multiple client engagements.

cFive Solutions – as a Senior Full-Stack Software Engineer *Laguna Hills/CA, Jun/2018 – Mar/2019*

Designed and delivered secure, enterprise-grade authentication solutions for government agencies, focusing on identity management, access control, and high-availability distributed systems. Architected scalable security components, modernized legacy authentication flows, and ensured compliance with federal security standards.

Key Projects & Contributions:

- **SAML 2.0 Security Platform** – Architected and developed a security platform using Java, Spring Boot, and Spring Security SAML to enable secure, federated authentication for the Los Angeles County main system, accessed by agencies such as the FBI and local law enforcement.
- **Microservices-Based Authentication** – Designed authentication model leveraging Microsoft AAD/MIM for identity management, ensuring high availability, scalability, and strict security compliance for government environments.
- **Spring Boot Service Provider for SAML 2.0** – Implemented fully annotation-driven service provider (no XML), improving maintainability and eliminating legacy configuration overhead.
- **End-to-End Authentication Flows** – Built comprehensive authentication flows integrating SSO, metadata handling, certificate validation, and secure token exchange for cross-agency access.
- **Federal Compliance Collaboration** – Collaborated with security teams and government stakeholders to align the solution with federal authentication and privacy standards.

Impact: Engineered a mission-critical authentication architecture adopted across multiple government agencies, significantly improving security, interoperability, and cross-department access while modernizing legacy authentication workflows.

Conversant – as a Software Engineer

Westlake Village/CA, Jun/2017 – Jan/2018

Delivered backend engineering for a high-traffic Ad Network SaaS platform, modernizing legacy systems, building scalable APIs, and implementing distributed architecture solutions. Focused on feature development, system reliability, and operational efficiency using Java/Spring Boot and cloud-native patterns.

Key Projects & Contributions:

- **RESTful API Development** – Developed and maintained Java/Spring Boot RESTful APIs for an online Ad Network platform, delivering new SaaS features, modernizing legacy components, and resolving bugs using TDD and Agile practices.
- **Spring Cloud Config Server** – Built configuration server from scratch, enabling centralized, version-controlled configuration management across development, staging, and production environments. Configuration updates could be applied dynamically without redeployment, improving operational efficiency and reducing downtime.
- **Distributed Caching Solution** – Implemented Hazelcast-based caching with Spring AOP, preventing concurrent processing of correlated service requests across multiple servers and enhancing system consistency and throughput.
- **Cross-Functional Collaboration** – Worked with cross-functional teams to ensure high-quality releases, improve system reliability, and accelerate feature delivery in a fast-paced SaaS environment.

Impact: Designed and implemented a distributed caching mechanism with Hazelcast that eliminated concurrency conflicts across multiple servers, increasing system throughput and improving user experience for high-traffic Ad Network operations. Delivered a zero-downtime, dynamic configuration architecture using Spring Cloud Config, significantly improving deployment efficiency and system reliability.

CERTIFICATES

- [CKAD - Certified *Kubernetes* Application Developer](#);
- [HashCorp Certified *Terraform* Associate](#);
- [AWS Certified *SysOps* Administrator](#);
- [AWS Certified *Developer* Associate](#);
- [CFCD - *Cloud Foundry* Certified Developer](#);
- [React \(by Udacity\)](#).

ACADEMIC QUALIFICATIONS

- **Bachelor of Technology in Data Processing**
Centro Universitário UNA, Brazil
- **Professional Specialization in Software Engineering**
Universidade Federal de Minas Gerais, Brazil

TECHNICAL SKILLS

- **AI/ML:** Python, LangChain, LangGraph, MCP, RAG Pipelines, Multi-Agent Systems, Generative AI
- **Backend:** Java, Spring Boot, Python, Microservices, RESTful APIs, Design Patterns
- **Cloud & Infrastructure:** AWS, Azure, Kubernetes, Docker, Terraform, Terragrunt, Helm, Istio
- **DevOps & Observability (SRE):** CI/CD, Grafana, Splunk, Dynatrace, Prometheus
- **Enterprise:** SAP BTP/CAP/CPI, Event-Driven Architecture, Camunda BPMN
- **Practices:** Agile/Scrum, TDD, DDD, Microservices & Microfrontend Architecture